<u>Lemma</u> Let (M, g) be compact, $|Rm| \leq K$,

$\mathrm{Vol}\, B(p,1) \geq v > 0 \quad \forall p \in M$.

Then $\mathrm{injrad}\, M \geq i_0$, i_0 dep. on n, K, v

<u>Proof</u> By contradiction: suppose (M_i, g_i) satisfies $|Rm| \leq K$, $\mathrm{vol}\, B(p,1) \geq v \quad \forall p$

and $p_i \in M_i$ satisfy

$$\mathrm{injrad}\, p_i = \mathrm{injrad}\, M_i = \delta_i \longrightarrow 0$$

Consider rescaled manifolds

$$(\tilde{M}_i, \tilde{g}_i, \tilde{p}_i) \qquad \begin{aligned} \tilde{M}_i &= M_i \\ \tilde{g}_i &= \delta_i^{-2} g_i \\ \tilde{p}_i &= p_i \end{aligned}$$

$$\mathrm{injrad}\, \tilde{M}_i = \mathrm{injrad}\, \tilde{p}_i = 1$$

$$|Rm(\tilde{g}_i)| \leq \delta_i^2 K \longrightarrow 0$$

$\Longrightarrow$ From previous lemma we can choose a subsequence converging to

$(\tilde{M}, \tilde{g}, \tilde{p})$ in pointed $C^{1,\alpha}$-sense

and $\|(\tilde{M}, \tilde{g})\|_{C^{1,\alpha}, 1/2} = 0$

i.e. $(\tilde{M}, \tilde{g})$ is flat.

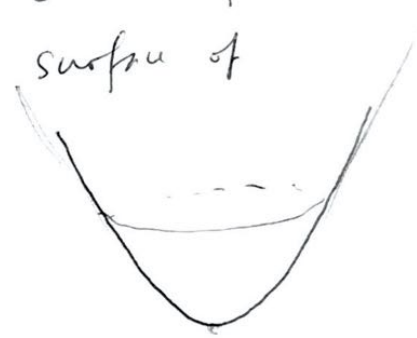
Contradiction arises from the following:

1) injrad $\tilde{M} = 1$

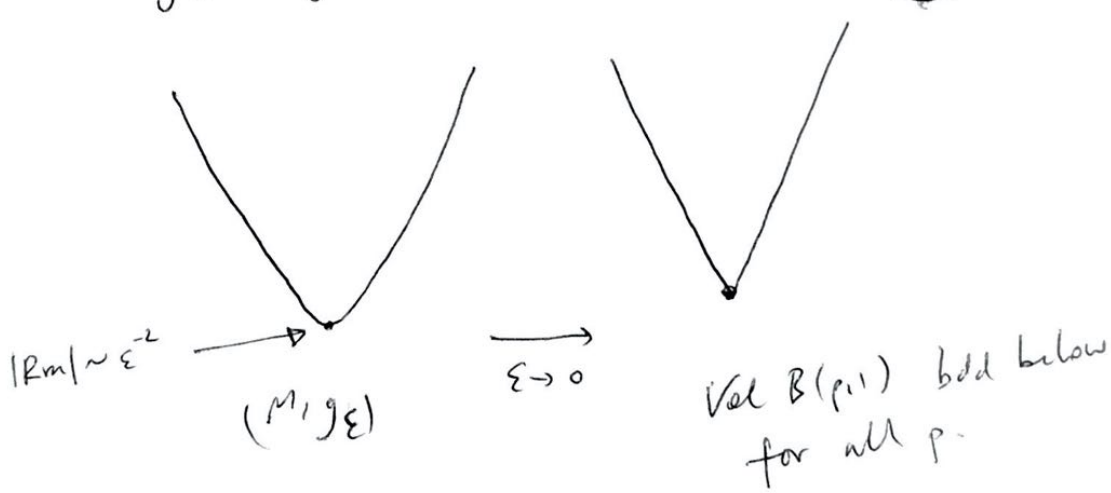
2) $\mathrm{Vol}\, B(\tilde{p}, r) \geq v' r^n$ for some $v' > 0$ and all $r > 0$

Only flat mfld satisfying (2) is $\mathbb{R}^n$ which does not satisfy (1). $\square$

<u>E.g.</u> Not true without upper bound for curvature: let (M, g) be surface of revolution of $y = \sqrt{1 + x^2}$



let $g_\varepsilon = \varepsilon^2 g$



Harmonic coordinates

Results so far are in the spirit of estimates like

$$\text{if } |\nabla^2 f| \leq C \Rightarrow |f|_{C^\alpha} \leq C'$$

since Rm ~ Hessian of metric

We want estimates like Schauder estimates

$$|\Delta f|_{C^\alpha} \leq C \Rightarrow |f|_{C^{2,\alpha}} \leq C'$$

Intuition: Ric ~ Laplacian of g.

<u>Recall</u>: $u : M \to \mathbb{R}$ is harmonic if $\Delta u = 0$

- in coordinates $\Delta u = g^{jk} \partial_j \partial_k u - g^{jk} \Gamma^{\ell}_{jk} \partial_\ell u$
- e.g. if M is Kähler then in holo. coordinates $\Delta u = g^{j\bar{k}} \partial_j \partial_{\bar{k}} u$ and if $f : M \to \mathbb{C}$ is holomorphic then $\operatorname{Re} f$, $\operatorname{Im} f$ are harmonic

<u>Definition</u> chart $x = (x^1, \dots, x^n) : B(p,r) \longrightarrow \mathbb{R}^n$ is a harmonic coordinate system if $\Delta x^i = 0 \quad \forall i$.

<u>E.g.</u> if M is Kähler, and $z^1, \dots, z^n$ are holo. coords, then $(x^1, \dots, x^n, y^1, \dots, y^n)$ is a harmonic coord system, $z^i = x^i + \sqrt{-1} y^i$

<u>Useful facts:</u> suppose $\{x^i\}$ are harmonic coords.

- since $\Delta x^k = \cancel{g^{ij}\partial_i\partial_j x^k} - g^{ij}\Gamma^l_{ij}\underbrace{\partial_l x^k}_{\delta_l^k}$

$$\Rightarrow \quad g^{ij}\Gamma^k_{ij} = 0 \quad \forall k$$

so for any $u : M \to \mathbb{R}$ we have

$$\Delta u = g^{ij}\partial_i\partial_j u$$

- Ricci curvature in harmonic coordinates:

$$Ric_{ij} = -\frac{1}{2}\Delta g_{ij} + \underbrace{Q_{ij}(g, \partial g)}_{\text{non-linear expression, quadratic in } \partial g} \qquad (*)$$

<u>e.g.</u> in holomorphic coordinates on Kähler mfld

$$R_{i\bar{j}k\bar{l}} = -\partial_k\partial_{\bar{l}}g_{i\bar{j}} + g^{p\bar{q}}\partial_i g_{k\bar{q}}\partial_{\bar{j}}g_{p\bar{l}}$$

$$Ric_{i\bar{j}} = g^{k\bar{l}}R_{i\bar{j}k\bar{l}} = -\Delta g_{i\bar{j}} + \underbrace{g^{k\bar{l}}g^{p\bar{q}}\partial_i g_{k\bar{q}}\partial_{\bar{j}}g_{p\bar{l}}}_{Q(g,\partial g)}$$

In Riemannian case $(*)$ follows from Bochner formula:

$$\frac{1}{2}\Delta|\nabla u|^2 = |\nabla^2 u|^2 + g(\nabla u, \nabla\Delta u) + Ric(\nabla u, \nabla u)$$

applied to $u = x^k$, using $\Delta x^k = 0$:

$$\frac{1}{2}\Delta |\nabla x^k|^2 = |\nabla^2 x^k|^2 + \mathrm{Ric}(\nabla x^k, \nabla x^k)$$

polarize this (apply to $x^k \pm x^\ell$):

$$\underbrace{\mathrm{Ric}(\nabla x^k, \nabla x^\ell)}_{g^{kp} g^{\ell q} \mathrm{Ric}_{pq}} = \frac{1}{2}\Delta \underbrace{g(\nabla x^k, \nabla x^\ell)}_{g^{k\ell}} - g(\nabla^2 x^k, \nabla^2 x^\ell)$$

<u>Proposition</u> $\exists\, Q > 0$ (small) such that if

$\|(M, g, p)\|_{C^{m,\alpha}, 1} < Q$ $(m \geq 1)$, then

$\exists$ harmonic coords on $B(p, 1/2)$ s.t.

metric components satisfy

$$\frac{1}{2}\delta_{ij} < g_{ij} < 2\delta_{ij}$$

$$\|g_{ij}\|_{C^{m,\alpha}} \leq 2$$

<u>Proof</u> we have a coordinate map

$$\varphi : B(0,1) \longrightarrow M, \qquad \varphi(0) = p$$

s.t. $$e^{-2Q}\delta_{ij} \leq (\varphi^* g)_{ij} \leq e^{2Q}\delta_{ij}$$

$$\|g_{ij}\|_{C^{m,\alpha}} \leq Q$$

Let x^i be Euclidean coords on $B(0,1)$

Define y^i by $\Delta y^i = 0$ on $B(0, 3/4)$

g-Laplacian $\quad y^i = x^i$ on $\partial B(0, 3/4)$

$$\Delta x^i = g^{kl}\partial_k\partial_l x^i - g^{kl}\Gamma^p_{kl}\partial_p x^i$$
$$= -g^{kl}\Gamma^i_{kl}$$

$\Rightarrow \|\Delta x^i\|_{C^{m-1,\alpha}} \leq CQ$

max principle $\Rightarrow |x^i - y^i| \leq CQ$ on $B(0,3/4)$

Schauder estimates $\Rightarrow$

$$\|x^i - y^i\|_{C^{m+1,\alpha}(B(0,2/3))} \leq C\left(\|\Delta(x^i-y^i)\|_{C^{m-1,\alpha}(B(0,3/4))} + \|x^i-y^i\|_{C^0(B(0,3/4))}\right)$$
$$\leq CQ$$

$\Rightarrow$ If Q is suff. small, the y^i are coords on $B(0,2/3)$, and the metric components satisfy the required estimates. $\square$